Solving System of Equations using Matrices

4. Using algebraic elimination, find the point of intersection of these three planes.

$$\begin{aligned} x + y + z &= -1 \\ 2x + 2y + 3z &= -7 \\ 3x - 2y + 7z &= 4 \end{aligned}$$

5. Write the following linear systems in matrix form.

a. $\begin{aligned} 5x - 2y + z &= 5 \\ 3x + y - 5z &= 12 \\ x - 5y + 2z &= -3 \end{aligned}$ b. $\begin{aligned} -2x + y - 3z &= 0 \\ x + 5y &= 8 \\ 3y + 2z &= -6 \end{aligned}$ c. $\begin{aligned} 4y - 3z &= 12 \\ 2x + 5y &= 15 \\ 4x + 6z &= 10 \end{aligned}$

6. Write the equations that correspond to the following matrices.

a. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & 3 \end{array}\right]$ b. $\left[\begin{array}{ccc|c} 1 & 0 & -6 & 4 \\ 0 & 1 & 5 & -5 \\ 0 & 0 & 0 & 0 \end{array}\right]$ c. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$

8. Using Gaussian elimination, state whether the system has no solutions, a unique solution, or an infinite number of solutions.

a) $\begin{aligned} x - y + z &= 5 \\ -x + y + 2z &= 7 \\ 2x + y - z &= 4 \end{aligned}$ b) $\begin{aligned} 2x + y - 3z &= 1 \\ -6x - 3y + 9z &= 3 \\ 4x + 2y - 6z &= 2 \end{aligned}$ c) $\begin{aligned} x - 2y + z &= 1 \\ -3x - 6y + 3z &= -3 \\ 2x - 4y + 2z &= 4 \end{aligned}$

d) $\begin{aligned} 4x - 3y - 15z &= 20 \\ x + 6y + 3z &= 5 \\ 2x - y - 7z &= 10 \end{aligned}$ e) $\begin{aligned} 3x + 5y - z &= 1 \\ -6x - 10y + 2z &= -2 \\ 12x + 20y - 4z &= 4 \end{aligned}$ f) $\begin{aligned} x - 3y &= 1 \\ -2x + 4y + z &= 1 \\ -2y + z &= 3 \end{aligned}$

Answers

4. $\left(\frac{47}{5}, \frac{-27}{5}, -5\right)$

5. a. $\left[\begin{array}{ccc|c} 5 & -2 & 1 & 5 \\ 3 & 1 & -5 & 12 \\ 1 & -5 & 2 & -3 \end{array}\right]$ **b.** $\left[\begin{array}{ccc|c} -2 & 1 & -3 & 0 \\ 1 & 5 & 0 & 8 \\ 0 & 3 & 2 & -6 \end{array}\right]$

c. $\left[\begin{array}{ccc|c} 0 & 4 & -3 & 12 \\ 2 & 5 & 0 & 15 \\ 4 & 0 & 6 & 10 \end{array}\right]$ **6. a.** $x = 8, y = -6, z = 3$

b. $x - 6z = 4, y + 5z = -5, 0z = 0$ **c.** $x = 0, y = 0, 0z = 1$

8 a) Three non-parallel planes meet at a single point $(3,2,4)$

b) No solution: 2 coincident planes, the third is parallel and distinct

c) No solution: 2 parallel and distinct planes, the third is non-parallel and intersecting both in two different lines

d) Infinite # of solutions: Three non-parallel planes (star shape) intersect in the line $\vec{r} = (5,0,0) + t(3,-1,1), t \in R$

e) Infinite # of solutions: Three coincident planes meeting in a plane of intersection $3x + 5y - z = 1$

f) Infinite # of solutions: Three non-parallel planes (star shape) intersect in the line $\vec{r} = \left(\frac{-7}{2}, \frac{-3}{2}, 0\right) + t(3,1,2), t \in R$